

Ex1b: Hidden Markov Model (HMM) based Predictive Text System

LAB EXPERIMENT x GENAI1b.ipynb x Copy of Untitled: x 1b.pdf - Google x No Code in PDF x Inbox (2,738) - x DOC-20260106- x Digicampus x Convert Word to x Home - Google x +

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Rajalakshmi Engine...

GENAI1b.ipynb

File Edit View Insert Runtime Tools Help

Commands + Code + Text Run all

```
[1] ✓
import nltk
nltk.download('brown')
nltk.download('universal_tagset')
```

[nltk_data] Downloading package brown to /root/nltk_data...
[nltk_data] Unzipping corpora/brown.zip.
[nltk_data] Downloading package universal_tagset to /root/nltk_data...
[nltk_data] Unzipping taggers/universal_tagset.zip.
True

```
[1] ✓
from nltk.corpus import brown
sentences=brown.sents()
tokens=[word.lower() for sent in sentences for word in sent]
print("Total sentences",len(sentences))
print("Total tokens",len(tokens))
tokens[:20]
```

Total sentences 57340
Total tokens 1161192
['the',
 'fulton',
 'country',
 'grand',
 'jury',
 'said',
 'friday',
 'an',
 'investigation',
 'of',
 'atlanta's',
 'recent',
 'primary',
 'election',
 'produced',
 '...',
 'no',
 'evidence',
 '...']

Variables Terminal

8:55 AM Python 3

15°C Mostly clear

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print("Total tokens",len(tokens))
tokens[:20]
```

```
[1] ✓
from nltk.tag import hmm
from collections import defaultdict, Counter
train_data = brown.tagged_sents(tagset='universal')
trainer = hmm.HiddenMarkovModelTrainer()
hmm_model = trainer.train_supervised(train_data)
print("HMM model trained successfully.")
```

HMM model trained successfully.

```
[1] ✓
import math
pos_sequences = [[tag for (_, tag) in sent] for sent in train_data]
pos_bigram = defaultdict(Counter)
for seq in pos_sequences:
    for i in range(len(seq)-1):
        pos_bigram[seq[i]][seq[i+1]]+=1
word_by_tag = defaultdict(Counter)
for sent in train_data:
    for word, tag in sent:
        word_by_tag[tag][word.lower()] += 1
```

Variables Terminal

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LAB EXPERIMENT GENAI/b.pytb

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for seq in pos_sequences:
    for i in range(len(seq)-1):
        pos_bigram[seq[i]][seq[i+1]]+=1
word_by_tag = defaultdict(counter)
for sent in train_data:
    for word, tag in sent:
        word_by_tag[tag][word.lower()] += 1
S=5
def predict_next_word(sentence):
    words = sentence.lower().split()
    tagged = hmm_model.tag(words)
    last_tag = tagged[-1][1]
    next_pos = pos_bigram[last_tag].most_common(1)[0][0]
    candidates = word_by_tag[next_pos].most_common(S)
    return next_pos, candidates
pos, words = predict_next_word("the government decided")
print("Predictes POS:", pos)
print("Suggested Next Words:", words)

Predictes POS: NOUN
Suggested Next Words: [('time', 3597), ('man', 3283), ('af', 995), ('years', 949), ('way', 899)]
```

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